PROJECT DOCUMENTATION / SECTION 04

UI/UX Design and Prototyping

Design principles, interface standards, interaction behaviour, responsiveness, and accessibility for the SIEM SOC Operations Dashboard.

Purpose

This document defines the user-interface and user-experience approach for the SIEM SOC Operations Dashboard. It establishes consistent visual rules and interaction principles for the six primary screens, with an emphasis on readability, efficient alert review, and clear security status communication.

Project scope

The interface is designed to support detection engineering, SIEM visibility, alert investigation, operational monitoring, threat-intelligence review, and hand-off to an existing SOAR platform. Any values shown in the design are illustrative sample data unless a real integration is explicitly implemented and verified.

Prepared as the UI/UX documentation component of the project submission.

1. Wireframes and Mockups

Insert the approved wireframe and final mockup images under the corresponding headings below. This document intentionally reserves these sections for the project’s actual screen captures.

1.1 Detection Rules

Detection rule inventory, severity, ATT&CK; mapping, validation status, and example rule logic.

1.2 SIEM Integration

Log-source flow from collection and forwarding through Splunk, analyst triage, and the existing SOAR platform.

1.3 Alert Investigation

Selected alert context, evidence, activity timeline, analyst decision, and escalation summary.

1.4 Operations Dashboard

Operational KPIs, alert-volume trend, source and target summaries, and key SOC performance indicators.

1.5 Threat Intelligence

IOC enrichment, confidence and reputation indicators, and ATT&CK; technique distribution.

1.6 SOAR Integration

Response workflow, example playbooks, analyst approval points, and integration guardrails.

SOC OPERATIONS

SIEM / DETECTION ENGINEERING

LAB VIEW - SAMPLE DATA

WORKSPACE

01 Detection Rules

02 SIEM Integration

03 Alert Investigation

04 Operations Dashboard

05 Threat Intelligence

06 SOAR Integration

ENVIRONMENT

Production-like lab
Splunk Enterprise

Connected in concept

Detection Rules

Correlation logic, severity, MITRE ATT&CK mappings, and tuning status.

ACTIVE RULES	NEEDS TUNING	COVERAGE	LAST VALIDATION
08	02	06	Today
Insterative rules in this prototype	Review false-positive rate	Technique groups represented	Sample test attacks only

RULE / USE CASE	SEVERITY	ATT&CK	STATUS	LAST TEST
Brute-force login	HIGH	T1110	Enabled	Passed
Suspicious successful login	HIGH	T1078	Enabled	Passed
Lateral movement indicators	CRITICAL	T1021	Tuning	Review
Persistence via scheduled task	HIGH	T1053	Enabled	Passed
Possible data exfiltration	CRITICAL	T1041	Enabled	Passed
Endpoint malware alert	HIGH	T1084	Enabled	Passed

Example logic - Brute-force authentication

Trigger when failed logins exceed a tuned threshold for one account or source within a time window, then correlate any subsequent success.
Exclude approved scanners and known test accounts.

Tune by baseline, asset criticality, identity context, and validated test cases. Values shown are illustrative.

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SIEM Integration

Log sources are forwarded and normalized in Splunk before detections create alerts for analyst triage.

LOG SOURCES	→	COLLECT & FORWARD	→	SPLUNK SIEM	→	ANALYST TRIAGE	→	EXISTING SOAR
Endpoint · AV Firewall · IDS/IPS Identity · Cloud · Apps		UF / syslog / APIs Transport & parsing		Normalize fields Search · Correlate Generate alerts		Validate evidence Enrich context Assign severity		Approved playbooks Contain · notify Record actions

AUXILIARY ENVIRONMENT

Threat Intelligence (IOC reputation + MITRE ATT&CK context) • Ollama: analyst-assist summaries, investigation prompts, draft reports

Implementation notes

- Splunk is the only SIEM shown in this architecture.
- Existing SOAR is integrated; it is not being rebuilt as part of this prototype.
- Data is illustrative. No live telemetry or working connector is implied by this design.

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Alert Investigation

A repeatable triage workflow for validating alerts, enriching evidence, and escalating when needed.

SELECTED ALERT

Multiple failed logins followed by a successful authentication

Alert ID: SAMPLE-1042 | Source: Identity logs | Severity: High | State: New

Sample sensor – not a live Splunk alert

WORKFLOW STATE

1 Validate

2 Enrich

3 Decide & document

Evidence & context

Identity	User / role / MFA status
Source	IP reputation and geo context
Asset	Criticality and owner
Timeline	Failures → success → follow-on activity

ALERT ACTIVITY TIMELINE

Related signals

Failed logins · Successful login · Endpoint signal · Analyst review

Analyst decision

Benign / expected

Suspicious – investigate

Confirmed incident – escalate

Record rationale and evidence before closure.

Investigation notes / escalation summary

What happened • evidence observed • scope and affected assets • actions taken • remaining uncertainty • recommnnmended next step

Prepare escalation

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Operations Dashboard

SOC KPIs and queue health. All figures are sample values for the design prototype.

MTTD
Mean time to detect
4m 32s
↓ 28% vs last week

MTTR
Mean time to respond
18m 10s
↓ 32% vs last week

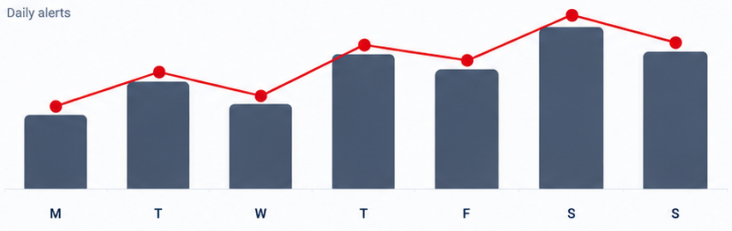
MTTC
Mean time to contain
11m 05s
↓ 25% vs last week

Dwell Time
Average Dwell Time
26m
↓ 41% vs last week

ALERT VOLUME	CLOSURE RATE	OPEN ALERTS	ESCALATED
248	86%	34	07
Alerts in selected period	Closed / total in period	Current sample queue	Requiring higher-tier review

Alert volume - sample 7-day trend

Daily alerts



Top source IPs - sample

198.51.100.24	42
203.0.113.17	27
192.0.2.60	19

Top targeted assets - sample

vpn-gateway-01	31
ad-server-02	24
web-prod-03	18

KPI definitions

MTTD: event-to-detection • MTTR: alert-to-response • MTTC: alert-to-containment • Dwell time: estimated time from initial compromise to detection/containment

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Threat Intelligence

Enrich indicators and map detections to ATT&CK. Data below is illustrative.

IOC RECORDS	HIGH CONFIDENCE	ATTACK TECHNIQUES	LAST SYNC
126	18	14	Sample
Sample enrichment set	Requires analyst review	Mapped in this sample	No live feed configured

INDICATOR	TYPE	REPUTATION	CONFIDENCE	ACTION
198.51.100.24	IPv4	Suspicious	High	Investigate
update-check.example	Domain	Unknown	Low	Monitor
a4f1...9b2c	File hash	Malicious*	High	Escalate
203.0.113.17	IPv4	Suspicious	Medium	Enrich

*Example indicator. Hashes are placehoders, removed documentation of usage and example domains are not real threat attribution.

ATT&CK technique distribution

High 75%

Low 45%

Initial Access	Credential Access	Lateral Movement	Persistence	Exfiltration
T1078	T1110	T1021	T1053	T1041

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SOAR Integration

Handoff from Splunk alerts to an existing SOAR platform and approved response playbooks.

Splunk alert Detective fires → **Enrichment** IOC + asset context → **SOAR playbook** Existing platform → **Analyst approval** Where required → **Response** Contain / notify

Example response playbooks

Phishing response	Collect message, enrich URLs/redirects, isolate only with approval
Malware response	Collect endpoint evidence, validate severity, contain host per policy
Account compromise	Revoke sessions, reset credentials, notify MFA, notify identity team

PLAYBOOK HEALTH - SAMPLE

Phishing 90%

Malware 75%

Account compromise 60%

Ollama - analyst assistance

- Summarize alert context
- Suggest investigation questions
- Draft an incident report
- Keep analyst validation in the loop

AI output is advisory. No autonomous action is implied.

Integration guardrails

Use approved API/network integrations, least-privilege credentials, audit logging, alerts, and analyst approval for disruptive actions. This is a design concept, not a live integration.

2. UI/UX Guidelines

The interface follows a restrained, information-first design approach suited to a security operations environment.

2.1 Visual hierarchy

Use a clear page title, a short explanatory subtitle, and a predictable order for the main content. Place high-priority indicators first, followed by trends and supporting detail. Group related information in cards or panels with consistent spacing and alignment.

2.2 Colour and status communication

The palette combines deep navy for navigation and primary structure with white and very light neutral surfaces for reading. Red is reserved for critical emphasis, active navigation, and high-risk states. Muted blue tones support secondary data and visual comparison. Colour must reinforce labels and icons rather than replace them.

- **Primary navy:** #003049 — navigation, headings, and key structural elements.
- **Slate:** #566178 — secondary text, chart series, and subdued controls.
- **Signal red:** #C1121F — critical emphasis, active states, and high-priority indicators.
- **White:** #FDFDFD — primary content surfaces and readable space.
- **Pale neutral:** #F0E9EF — subtle supporting surface where needed.
- **Light blue:** #A8DADC and **muted blue:** #669BBC — secondary series, supporting indicators, and non-critical chart elements.

2.3 Typography

Use a consistent sans-serif typeface such as Inter or a system equivalent. Page titles should be visually distinct from section headings; table headings and labels should remain compact but readable. Use sentence case for most interface copy, consistent terminology for SOC metrics, and avoid unnecessary all-caps text outside short labels.

2.4 Layout, spacing, and surfaces

Align cards and tables to a consistent grid. Use a repeatable spacing scale and maintain clear separation between sections. Cards should have consistent corner radii, border weight, internal padding, and heading placement. Alternate table-row surfaces subtly but visibly to support scanning without reducing contrast. Prevent text collisions by allowing content to wrap or by increasing component height.

2.5 Charts and KPI cards

KPI cards should present a short label, a prominent value, a clear definition or time context, and a restrained visual trend where relevant. Keep chart legends, axes, labels, and plotted data visually distinct. Use line overlays only where they convey a meaningful trend. Do not add charts to screens that do not need them.

2.6 Tables and data density

Use consistent column alignment and adequate row height. Highlight severity and workflow state with labelled badges or text in addition to colour. Use alternating row fills with enough contrast to distinguish adjacent records. Long indicators, titles, and notes must not overlap neighbouring columns.

2.7 Terminology and metric definitions

Use metric names consistently throughout the interface: MTTD means Mean Time to Detect; MTTR means Mean Time to Respond; MTTC means Mean Time to Contain; and Dwell Time represents the estimated interval an attacker remains undetected in an environment. Any metric calculations must be documented and aligned with the project’s actual event timestamps and definitions.

3. Design System and Component Standards

Shared visual rules keep the six screens coherent and reduce inconsistencies during implementation.

3.1 Navigation

Keep the left navigation in a fixed, deep-navy visual treatment on desktop. Use clear labels and a distinct active state with the project red. Maintain consistent icon size, label alignment, vertical spacing, and grouping. On narrow screens, the navigation may collapse into a drawer or compact menu.

3.2 Cards and panels

Use a consistent card structure: optional icon or label, primary value or heading, supporting text, and optional action. Borders should define boundaries without dominating the page. Apply one consistent radius and padding scale. Avoid decorative elements that compete with security information.

3.3 Severity and workflow states

Apply a consistent severity vocabulary—Critical, High, Medium, and Low—and a consistent workflow vocabulary such as New, In Progress, Investigating, Escalated, and Closed. Pair each colour with text, and preserve the same mapping across dashboard cards, tables, and investigation screens.

3.4 Interaction states

Interactive elements should have clearly distinguishable default, hover, focus, selected, disabled, and loading states where applicable. Keyboard focus must remain visible. Buttons should use action-oriented labels and provide a clear indication when an action is unavailable or requires confirmation.

3.5 Data presentation

Dates, times, IP addresses, event counts, and identifiers should use consistent formats. Align numeric values for comparison. Distinguish sample or placeholder records from verified telemetry, and do not present mock values as live security measurements.

3.6 Reusable components

Maintain reusable styles or components for KPI cards, table headers and rows, severity badges, status chips, chart legends, section headers, navigation items, and empty or warning states. Changes to shared components should be applied consistently across all screens.

4. Interactive Prototype and Navigation

The Figma prototype should demonstrate the principal analyst journey rather than only present static screens. Navigation labels should link to their corresponding screens, and key controls should provide an observable response when selected.

4.1 Primary navigation flow

- Detection Rules → inspect rule coverage, severity, ATT&CK; mapping, and tuning status.
- SIEM Integration → review log-source onboarding and the data path into Splunk.
- Alert Investigation → inspect an alert, review evidence, record an analyst decision, and prepare an escalation summary.
- Operations Dashboard → review SOC performance indicators and alert trends.
- Threat Intelligence → review example indicators, confidence, reputation, and technique mappings.
- SOAR Integration → review the existing response workflow, playbook examples, approval points, and guardrails.

4.2 Expected prototype interactions

- Sidebar items navigate to the matching screen and clearly indicate the active destination.
- A selected alert or View Details action opens the related investigation view or a clearly defined detail state.
- Filters and date-range controls, if shown, visibly reflect the selected value.
- Analyst decision controls visibly distinguish the selected option; disruptive response actions require confirmation in a real implementation.
- View All and similar links lead to a relevant expanded list or destination, rather than remaining decorative.

4.3 Prototype boundary

A Figma prototype demonstrates intended interface behaviour; it does not establish a working SIEM connection, a functioning SOAR action, or live threat-intelligence enrichment. These capabilities must be verified separately during implementation and testing.

5. Responsive Design

The desktop dashboard is the primary workspace because SOC analysis often depends on comparing multiple data sources. Responsive behaviour should preserve the hierarchy and usability of the same information on smaller screens.

5.1 Desktop

- Use a persistent sidebar and a grid layout for KPI cards, charts, and data tables.
- Keep chart legends and table headers visible and aligned with their content.
- Use the available width for investigation context without stretching text lines excessively.

5.2 Tablet

- Reduce the number of cards per row and allow panels to stack when needed.
- Keep primary navigation accessible and avoid compressing charts to the point that labels become unreadable.
- Allow wide tables to scroll horizontally when a compact column layout would harm readability.

5.3 Mobile

- Collapse the sidebar into a menu or drawer and retain a clear page title.
- Stack KPI cards vertically or in a compact grid, keeping labels and values legible.
- Allow charts to resize and preserve their legend, axis labels, and meaning.
- Prioritise the most important alert fields; use a deliberate horizontal scroll or a record-detail layout for dense tables.

5.4 Responsive validation

Review each screen at representative desktop, tablet, and mobile widths. Confirm that text does not overlap, controls remain reachable, table rows remain understandable, and no essential status or action disappears.

6. Accessibility and Usability

6.1 Contrast and readability

Maintain sufficient contrast between text and its background, particularly for small table labels, muted descriptions, and coloured status badges. Avoid relying on faint borders or colour alone to distinguish important information.

6.2 Keyboard and focus

All interactive controls in the implemented interface should be reachable by keyboard, follow a logical focus order, and display a visible focus indicator. Avoid interactions that require a mouse-only gesture.

6.3 Clear labels and feedback

Use concise, unambiguous labels. Explain acronyms when they first appear in a user-facing context or provide a consistent glossary. Provide clear feedback after an action, and show a confirmation step before any action that could isolate a host, disable an account, or trigger another disruptive response.

6.4 Error, empty, and loading states

Design deliberate states for unavailable data, failed requests, loading content, empty search results, and permission restrictions. Distinguish an unavailable integration from a healthy integration and from an integration that has not been configured.

6.5 Usability review

Ask representative users to complete core tasks: locate a high-severity alert, interpret the KPI definitions, review the evidence, identify the relevant detection rule, and find the SOAR hand-off. Record where users hesitate or misunderstand a status, then refine labels, spacing, and navigation accordingly.

7. Data Integrity and Implementation Handoff

The interface must clearly distinguish design examples from operational telemetry. Use an explicit sample-data label wherever records, counts, trend lines, IP addresses, hashes, or timestamps are illustrative.

7.1 SIEM and SOAR boundaries

Splunk is the SIEM represented in the architecture. The SOAR platform is treated as an existing system to integrate with, not as a new platform to rebuild. The design should not imply that an API, webhook, feed, or automated response is connected unless that integration has been implemented and tested.

7.2 Handoff requirements

- Provide the Figma file link and identify the approved frames for each screen.
- Document the colour palette, typography, spacing, card styles, table styles, status colours, and interaction states.
- Identify which buttons and navigation elements are prototype-only and which have a confirmed implementation.
- Document metric definitions, sample-data assumptions, and any unresolved design decisions.
- Keep screen names and terminology consistent across the Figma file, the PDF, and the implementation documentation.

7.3 Acceptance criteria

- All six screens use the same navigation system and design language.
- Text and components do not overlap at the reviewed viewport sizes.
- Alternating table rows are visibly distinct while preserving readable contrast.
- KPI values, chart labels, legends, and status indicators are legible and consistently styled.
- The prototype demonstrates the main navigation and agreed analyst interactions.
- Sample data and unconfigured integrations are clearly identified.

End of Section 04 — UI/UX Design and Prototyping